

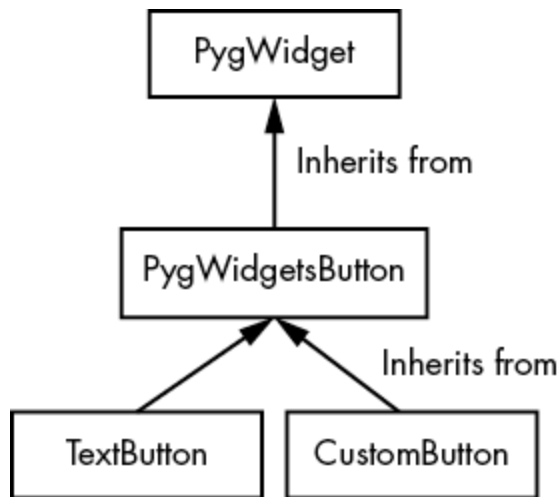


Figure 10-11: Class hierarchy in pygwidgets

As you can see, the `PygWidgetsButton` class is both a subclass of `PygWidget` and a base class for `TextButton` and `CustomButton`.

The Difficulty of Programming with Inheritance

When developing using inheritance, it can be difficult to understand what to put where. You're constantly asking yourself questions like: Should this instance variable be in the base class? Is there enough common code in subclasses to create a method in the base class? What are appropriate parameters for a method in a subclass? What are appropriate parameters and defaults to be used in a base class that expects to be overridden or called from a subclass?

Attempting to understand the interactions among all the variables and methods in a hierarchy of classes can be an extremely difficult, tricky, and frustrating task. This is especially true when reading the code of a class hierarchy developed by another programmer. To fully understand what's going on, you often have to become familiar with the code in the base classes all the way up the hierarchy.

For example, imagine a hierarchy in which class `D` is a subclass of `C`, which is a subclass of `B`, which is a subclass of the base class `A`. In class `D`, you may encounter code that branches based on the value of an instance